Jie Wu

PhD Student in Physics | Gravitational-Wave Physics
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About me

I am a PhD student in Physics at Chongqing University and expect to graduate in June 2028. My research focuses on gravitational-wave signals from moving compact binaries in hierarchical and astrophysical environments, particularly the effects of source motion, external gravitational fields, and environmental interactions on long-duration signals observed by space-based detectors. My work spans source dynamics, waveform calculations, detector response, and parameter estimation, with additional interests in tests of general relativity.

Research Interests

GW modeling	Waveform modeling, signal simulation, moving-source and environmental effects
Data analysis	Parameter estimation, Fisher-matrix analysis, signal processing
GW detectors	Space- and ground-based detectors, detector response, time-delay interferometry, multiband observations
Compact binary	Binary black holes, hierarchical triples, AGN disks, globular clusters
Tests of gravity	GW polarizations, parametrized tests of GR, modified gravity

Education

Chongqing University — *Chongqing, China*
PhD in Physics — *Sep 2022 - Jun 2028 (expected)*
- Advisor: Prof. Jin Li
- Research: GW modeling, moving-source and environmental effects, detector response, and parameter estimation

China West Normal University — *Nanchong, China*
B.S. in Physics — *Sep 2018 - Jun 2022*
- Advisors: Assoc. Prof. Di Wu and Assoc. Prof. Guo-Ping Li
- Research: Ground-based gravitational-wave detection and data analysis
- Thesis (in Chinese): An Analysis of the LIGO Gravitational Waves Data Based on Newtonian Approximate Model

Funding

7 funded research projects, including 3 led as Principal Investigator and 4 collaborative projects.

Signal Processing and Theoretical Modeling of Space-Based Gravitational Waves with Moving-Source Effects — *Principal Investigator*, *Jan 2026 – Dec 2027*
Young Students' Basic Research Program (PhD Students), National Natural Science Foundation of China
- Grant No. 125B2102 · CNY 300,000

Young Science and Technology Talent Cultivation Project – Doctoral Student Special Program — *Principal Investigator*, *Jan 2026 – Dec 2027*
China Association for Science and Technology
- CNY 40,000

Dark Matter and Modified Gravity — *Principal Investigator*, *Sep 2020 – Sep 2021*
Innovative Practice Training Program for College Students, University of Chinese Academy of Sciences
- Grant No. 20204001217 · CNY 5,000

Machine Learning-Optimized Multi-type Compact Binary Coalescence Gravitational Wave Waveform Generation — *Participant*, *Jan 2026 – Dec 2029*

General Program, National Natural Science Foundation of China

- Grant No. 12575072 · CNY 540,000

Simulation, Processing, and Validation of Gravitational-Wave Signals from Binary Black Hole Systems — *Participant*, *Aug 2024 – Dec 2028*

Subproject, National Key R&D Program of China

- Grant No. 2023YFC2206702 · CNY 950,000

Joint Observation of the Stochastic Gravitational-Wave Background from Cosmic Strings with Space-Based Detectors — *Participant*, *Jul 2023 – Jun 2026*

General Program, Natural Science Foundation of Chongqing, China

- Grant No. CSTB2023NSCQ-MSX0103 · CNY 100,000

Characterization and Signal Identification of Novel Gravitational-Wave Sources and Stochastic Gravitational-Wave Backgrounds, Including Cosmic Strings — *Participant*, *Jan 2022 – Sep 2026*

Subproject, National Key R&D Program of China

- Grant No. 2021YFC2203004 · CNY 1,100,000

Publications

16 publications, including 10 first-author papers.
Publications are listed in reverse chronological order (*: corresponding author; †: these authors contributed equally).

First-author

[1] **J. Wu**, Y. Xiao, M. Sun and J. Li*, Probing globular clusters using modulated gravitational waves from binary black holes, *Mon. Not. Roy. Astron. Soc.* accepted (2026), arXiv:2508.04021.

[2] **J. Wu**, M. Sun and J. Li*, Tests of general relativity using analytic derivatives of parametrized post-Einsteinian gravitational waveforms within the Fisher-matrix framework, *Eur. Phys. J. C* **86**, 1098 (2026), arXiv:2607.04238.

[3] **J. Wu**, M. Sun, J. Li* and Z. Cao*, Constraining AGN Disk Properties with Gravitational Waves from Inspiraling Stellar-Mass Binary Black Holes in Hierarchical Triple Systems, *Sci. China Phys. Mech. Astron.* accepted (2026), arXiv:2609.00945.

[4] **J. Wu**, J.-T. Yao, M. Sun, J. Li* and Z. Cao*, Constraining Kerr supermassive black hole properties using gravitational waves from inspiraling stellar-mass binary black holes, *Phys. Rev. D* **113**, 124024 (2026), arXiv:2606.12784.

[5] **J. Wu**, M. Sun and J. Li*, Constraints and detection capabilities of GW polarizations with space-based detectors in different TDI combinations, *Class. Quant. Grav.* **43**, 065017 (2026), arXiv:2411.03631.

[6] **J. Wu**, M. Sun, X. Ma, X. Liu, J. Li* and Z. Cao*, Effect of kick velocity on gravitational wave detection of binary black holes with space- and ground-based detectors, *Phys. Rev. D* **112**, 024040 (2025), arXiv:2502.13710.

[7] **J. Wu** and J. Li*, Prospects of constraining on the polarizations of gravitational waves from binary black holes using space- and ground-based detectors, *Phys. Rev. D* **110**, 084057 (2024), arXiv:2407.13590.

[8] **J. Wu**, J. Li*, X. Liu and Z. Cao, Comparison and application of different post-Newtonian models for inspiralling stellar-mass binary black holes with space-based GW detectors, *Phys. Rev. D* **109**, 104014 (2024), arXiv:2401.03113.

[9] **J. Wu** and J. Li*, Subtraction of the confusion foreground and parameter uncertainty of resolvable galactic binaries on the networks of space-based gravitational-wave detectors, *Phys. Rev. D* **108**, 124047 (2023), arXiv:2307.05568.

[10] **J. Wu**, J. Li* and Q.-Q. Jiang*, Application of Newtonian approximate model to LIGO gravitational wave data processing, *Chin. Phys. B* **32**, 090401 (2023).

Co-author

[11] M. Sun†, B. Wang†, **J. Wu**, J. Li* and S. Ye, Effects of solar wind plasma noise on stochastic gravitational wave background searches with a space based interferometer network, *Phys. Rev. D* **114**, 083507 (2026), arXiv:2607.05368.

[12] M. Sun, **J. Wu**, Q. Hu, J. Li*, N. Yang, X. Ma, B. Wang, M. Zhang and Y. Zhong*, Identifiability of g mode Resonances in Eccentric Binary Neutron Stars with Multidetector Observations, (2026), arXiv:2606.11959 (under review).

[13] M. Sun, **J. Wu**, Q. Hu, J. Li*, N. Yang, X. Ma, B. Wang, M. Zhang and Y. Zhong*, Detection of multiband lensed gravitational waves from dark matter halos with deep learning, *Phys. Rev. D* **113**, 103004 (2026), arXiv:2511.09107.

[14] M. Sun, **J. Wu**, J. Li*, B. McCane, N. Yang, X. Ma, B. Wang and M. Zhang, Conditional Autoencoder for Generating Binary Neutron Star Waveforms with Tidal and Precession Effects, *Phys. Rev. D* **112**, 084016 (2025), arXiv:2503.19512.

[15] X. Ma, B. Wang, N. Yang, J. Li*, B. McCane, M. Sun, **J. Wu**, M. Zhang and Y. Meng*, Identification of Stochastic Gravitational Wave Backgrounds from Cosmic String Using Machine Learning, *Phys. Rev. D* **112**, 064081 (2025), arXiv:2502.11804.

[16] Y. Hu, **J. Wu**, H. Luo, G. Su, X. Meng, L. Liu and G. Chen*, Parallel manipulation of multiple ink droplets via near-infrared light on lubricant infused surface, *Appl. Phys. Lett.* **126**, 021602 (2025).

Research Experience

Beijing Normal University *Beijing, China*

Visitor *Feb 2024 - Apr 2024*

- Advisor: Prof. Zhoujian Cao
- Research: Gravitational wave waveform simulation and moving source effect

University of Chinese Academy of Sciences *Beijing, China*

Undergraduate Researcher *Sep 2020 - Sep 2021*

- Advisor: Assoc. Prof. Yong Tang
- Research: Analysis of gravitational wave data
- Program: College Student Innovation and Practice Program

Honors & Awards

Awards

2025.12 **Advanced Individual in Scientific and Academic Innovation**, Chongqing U.
2023.12 **Second Prize (Ranked 2nd-3rd)**, The 7th Sichuan Chongqing Astronomy Competition
2022.06 **Excellent Graduation Thesis**, China West Normal U.
2022.05 **Outstanding Graduate**, China West Normal U.
2018.11 **Third Prize (Ranked 7th-8th)**, The 5th Sichuan Chongqing Astronomy Competition

Scholarships

2025.12 **National Scholarship**, Ministry of Education of China
2024.09 **Theoretical Physics Graduate Scholarship (Twice)**, Chongqing U.
2022-2023 **Graduate Academic Scholarship (Twice)**, Chongqing U.
2020-2022 **First-class Scholarship (Three times)**, China West Normal U.
2020.12 **Haotian Astronomy Scholarship**, Nanjing VasTech Astronomical Instrument & Equipment Co. Ltd.
2018-2021 **Second-class Scholarship (Four times)**, China West Normal U.

Skills

Programming Python, Mathematica, MATLAB
Data Analysis Parameter estimation, Fisher-matrix analysis, signal processing
Languages Chinese (native), English